

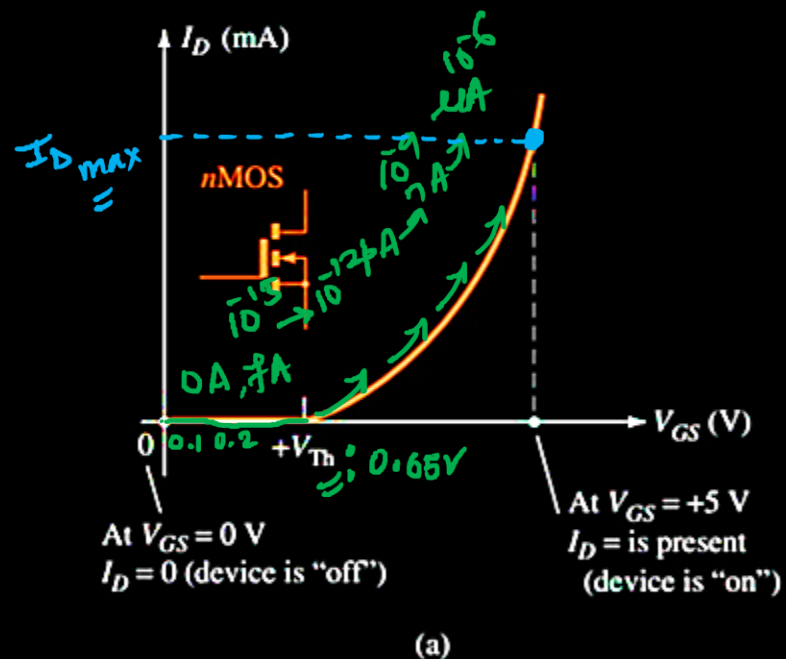
Anurag's Doubt

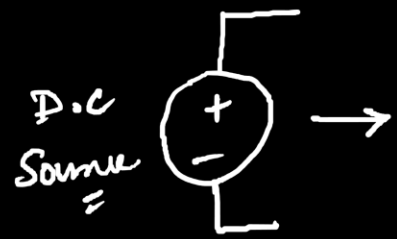
03-01-2026

pMOS On/Off Operation

The drain characteristic for a *p*-channel MOSFET or *p*MOS transistor is shown in Fig. 10.24b. When 0 V is applied, the device is "off" (no drain current present), whereas for an

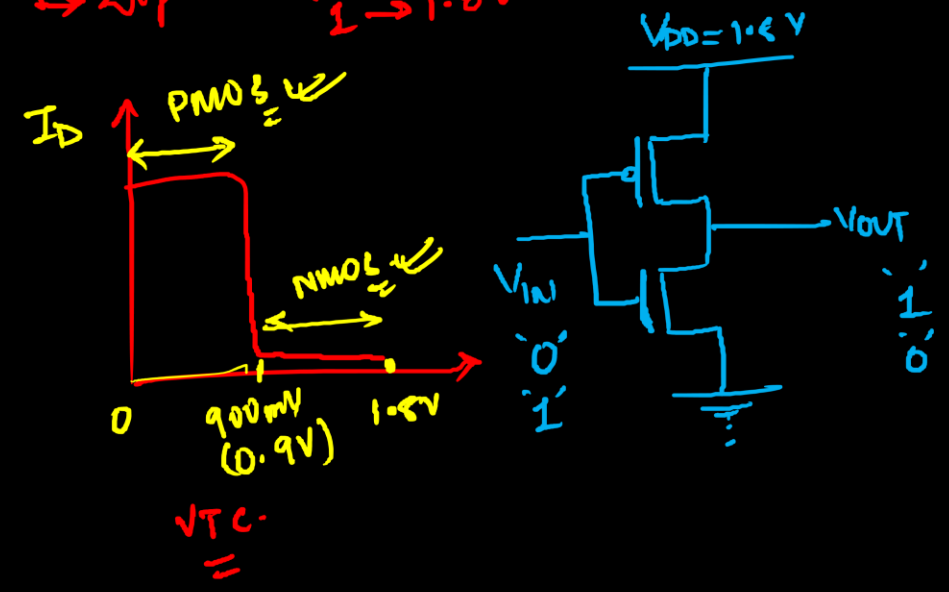
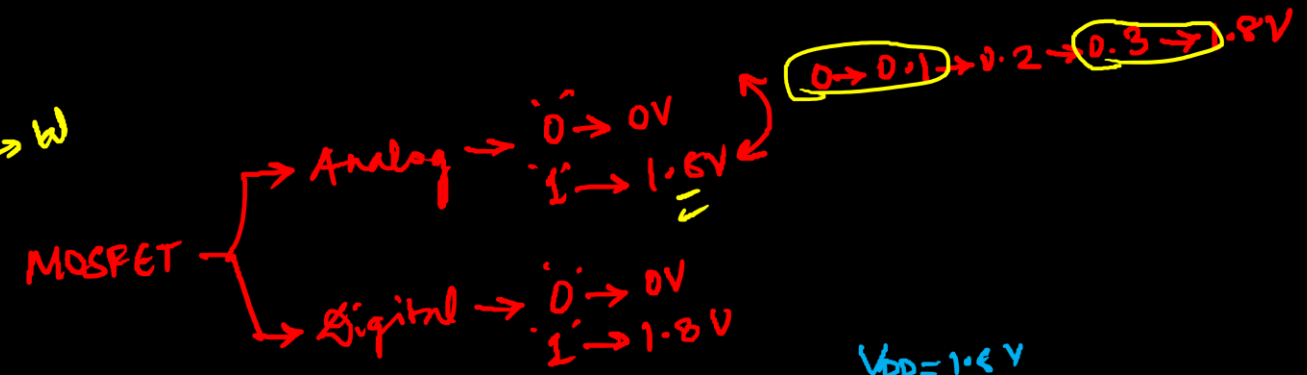
$I_D = I_S \left(e^{\frac{V_D}{V_T}} - 1 \right)$
 $BJT \rightarrow 0.7V_{BE}$
 -0.65
 $-10V$
 $-5V$
 $-4V$
 $-3V$
 $-2V$
 $-1V$
 0
 $1V$
 $2V$
 $3V$
 $4V$
 $5V$
 $V_{GS} \uparrow \uparrow \dots$
 $\times$ PMOS $\times$





Electrochemical cell. $(+ -) \rightarrow + \rightarrow -$: gradient (higher $\rightarrow$ lower)
 $\rightarrow$ emf $\rightarrow$ unit: "Newton"
 $\rightarrow$ V_{emf}

$(H) = I^2 R t$
 $\rightarrow$ In joules $\rightarrow$ mW $\rightarrow$ W

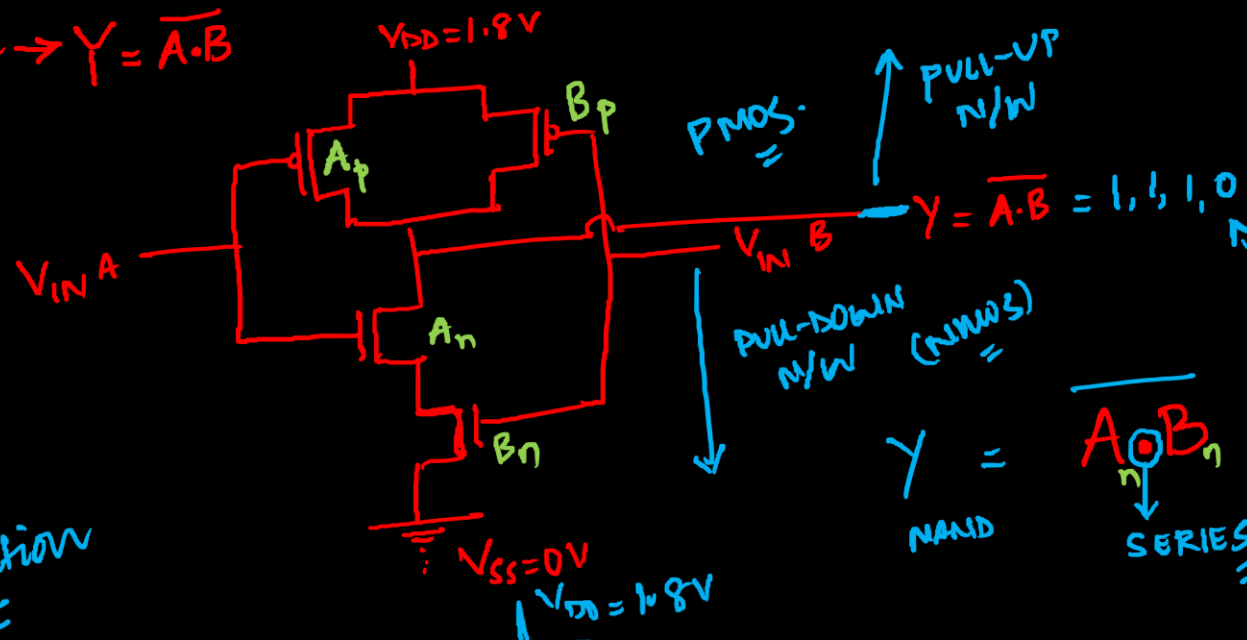


9V

- (+): 18V (-): 9V $= 18 - 9 = 9V$
- (+): -9 (-): -18 $= -(-18) + (-9) = +18 - 9 = +9V$

P.D $\rightarrow$ Potential difference $\rightarrow V_+ - V_- = V_N$

NAND gate $\rightarrow Y = \overline{A \cdot B}$



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

$$Y = \overline{A \cdot B} \quad \text{NAND}$$

SERIES

$$\overline{A \oplus B} = Y_{\text{NOR}}$$

PARALLEL

ONLY NAND for "NMOS"

